

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO2:** Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)  
Assessment Tool Weightage: 50%

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**QUESTIONS: 5**

**Total Marks:50**

Annexure B  
Analytical Problem Solving

| <i>Criteria</i>  | Problem Understanding<br>(2) | Method Selection<br>(2) | Solution Process<br>(2.5) | Accuracy of Calculation<br>(2) | Interpretation of Results (1.5) | <i>Total(10)</i> |
|------------------|------------------------------|-------------------------|---------------------------|--------------------------------|---------------------------------|------------------|
| <i>Weightage</i> | 20%                          | 20%                     | 25%                       | 20%                            | 15%                             | <i>100%</i>      |
| <i>Q1</i>        |                              |                         |                           |                                |                                 |                  |
| <i>Q2</i>        |                              |                         |                           |                                |                                 |                  |
| <i>Q3</i>        |                              |                         |                           |                                |                                 |                  |
| <i>Q4</i>        |                              |                         |                           |                                |                                 |                  |
| <i>Q5</i>        |                              |                         |                           |                                |                                 |                  |

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### ***Code of Conduct:***

*I \_\_\_\_\_Kamboji Thanuja\_192373012\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

*K Thanuja*

***Signature of the student:***

## RUBRICS FOR CALCULATION:

| Numerical Problems        |           |                                                                                   |                                           |                                            |                                               |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Criteria                  | Weightage | Level 4<br>(Exemplary)                                                            | Level 3<br>(Proficient)                   | Level 2<br>(Developing)                    | Level 1<br>(Beginning)                        |
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

CO-2 AT-1

1. Suppose that the data for analysis including the attribute age. The age values for the data tuples are (in increasing) order 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin depth of 3. Illustrate steps

a) What is the mean of the data? What is the median?

i) Mean ( $\bar{x}$ ) =  $\frac{1}{n} \sum_{i=1}^n x_i$

$N = 27$

$= 13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 + 35 + 35 + 35 + (36 + 40) + 45 + 46 + 52 + 70$

$= \frac{809}{27}$

$= 29.96$

approx = 30

ii) Median =  $\frac{(n+1)}{2}$

$= \frac{(27+1)}{2} = 14$

14th value is 25

b) What is the mode of the data? Comment on the data's modality (i.e., bimodal, Trimodal)

25 and 35 are repeating for 4 times

∴ It is Bimodal.



c) What is the midrange of the data?

$$\text{Midrange} = \frac{\text{Minimum} + \text{Maximum}}{2}$$

$$= \frac{13 + 70}{2} = 41.5$$

$$\text{Midrange} = 41.5$$

d) can you find (roughly) the first quartile ( $Q_1$ ) and the third quartile ( $Q_3$ ) of the data?

Lower half: ( $Q_1$ )

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$$\text{Median} = \frac{(13 + 1)}{2}$$

= 7<sup>th</sup> value of median is 20

Upper half: ( $Q_3$ )

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 50, 70

$$\text{Median value} = \frac{(13 + 1)}{2}$$

= 7<sup>th</sup> value of median is 35

Median is 35

$$\text{IQR} = Q_3 - Q_1$$

$$= 35 - 20$$

$$= 15$$

c. Give the five-number summary of the data.

Minimum = 13

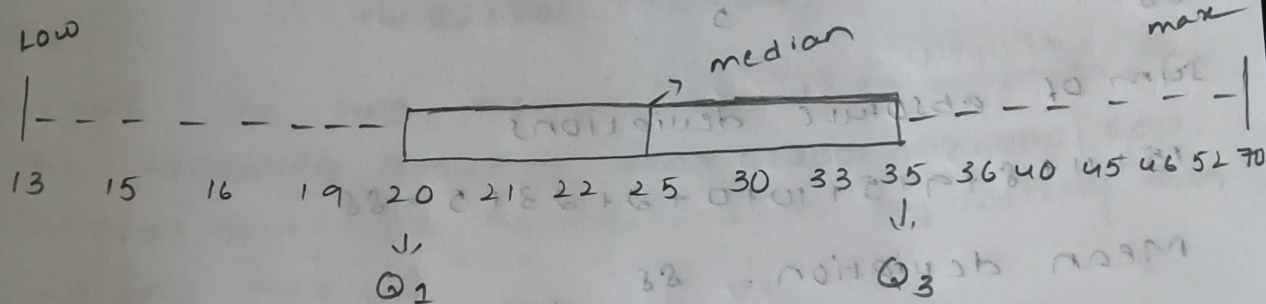
$Q_1 = 20$

Median = 25

$Q_3 = 35$

Maximum = 70

Show Box Plot of the data



2. Given the following measurements for the variables age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28 standardize the variable by following: compute the mean absolute deviation of age.

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Total} = 18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28$$

$$n = 10 \quad = 330$$

$$\bar{x} = \frac{330}{10} = 33$$



| <u>Age</u> | <u><math> x - \bar{x} </math></u> |
|------------|-----------------------------------|
| 18         | 15                                |
| 22         | 11                                |
| 25         | 8                                 |
| 42         | 9                                 |
| 28         | 5                                 |
| 43         | 10                                |
| 33         | 0                                 |
| 35         | 2                                 |
| 56         | 23                                |
| 28         | 5                                 |

Sum of absolute deviations

$$15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$$

$$\text{Mean deviation} = \frac{88}{10} = 8.8$$

$$= 8.8 \text{ years}$$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 50, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Total no. of observations : 27

Sum of observations = 809

Depth = 3

| <u>Bin</u>     | <u>values</u> | <u>mean</u> | <u>smoothed values.</u> |
|----------------|---------------|-------------|-------------------------|
| B <sub>1</sub> | 13, 15, 16    | 14.67       | 14.67, 14.67, 14.67     |
| B <sub>2</sub> | 16, 19, 20    | 18.33       | 18.33, 18.33, 18.33     |
| B <sub>3</sub> | 20, 21, 22    | 21          | 21, 21, 21              |
| B <sub>4</sub> | 22, 25, 25    | 24          | 24, 24, 24              |
| B <sub>5</sub> | 25, 25, 30    | 26.67       | 26.67, 26.67, 26.67     |
| B <sub>6</sub> | 33, 33, 35    | 33.67       | 33.67, 33.67, 33.67     |
| B <sub>7</sub> | 35, 35, 35    | 35          | 35, 35, 35              |
| B <sub>8</sub> | 36, 40, 45    | 40.33       | 40.33, 40.33, 40.33     |
| B <sub>9</sub> | 46, 52, 70    | 56          | 56, 56, 56              |

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1.

Min-max normalization (min = 0, max = 1)

$$\text{Normalized value} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

min value = 200

maximum = 1000

range = 800

max - min

1000 - 200

= 800

Original value

Normalized value

200

$$(200 - 200) / 800 = 0.00$$

300

$$(300 - 200) / 800 = 0.125$$

400

$$(400 - 200) / 800 = 0.250$$

600

$$(600 - 200) / 800 = 0.500$$

1000

$$(1000 - 200) / 800 = 1.00$$

includes  
for the  
13, 15,  
30, 33,  
smoothing  
at,  
steps.



## b) z-score Normalization

$$z = \frac{x - \mu}{\sigma}$$

Mean:

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

$$\mu = 500$$

Standard deviation

$$\sigma^2 = \frac{\sqrt{(200-500)^2 + (300-500)^2 + (400-500)^2 + (600-500)^2 + (1000-500)^2}}{5}$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000} \approx 282.84$$

| $x$  | $x - \mu$ | $(x - \mu)^2$ |
|------|-----------|---------------|
| 200  | -300      | 90,000        |
| 300  | -200      | 40,000        |
| 400  | -100      | 10,000        |
| 600  | 100       | 10,000        |
| 1000 | 500       | 250,000       |

| <u>original value</u> | <u>calculation</u>      | <u>z-score</u> |
|-----------------------|-------------------------|----------------|
| 200                   | $(200 - 500) / 282.84$  | -1.064         |
| 300                   | $(300 - 500) / 282.84$  | -0.71          |
| 400                   | $(400 - 500) / 282.84$  | -0.35          |
| 600                   | $(600 - 500) / 282.84$  | 0.35           |
| 1000                  | $(1000 - 500) / 282.84$ | 1.77           |



28/7/26

5. Suppose a group of 12 sales price records has been stored as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three e-bins by each of the following method.

a) equal-frequency partitioning.

b) equal-width partitioning

c) clustering.

a) equal-frequency partitioning

$$\frac{\text{Total}}{\text{No. of bins}}$$

Total NO. of records = 12

Bins = 3

$$= \frac{12}{3} = 4$$

Bin 1: 5, 10, 11, 13

Bin 2: 15, 35, 50, 55

Bin 3: 72, 92, 204, 215

b. equal-width partitioning

Width: Max - Min

No. of bins

Max = 215

min = 5

$$\frac{215 - 5}{3} = \frac{210}{3} = 70$$

So, intervals are

Bin 1: 5 - 74 → 5, 10, 11, 13, 15, 35, 50, 55, 72

Bin 2: 75 - 144 → 92

Bin 3: 145 - 215 → 204, 215

c. clustering

Group similar values together based on closeness.

A suitable clustering is

Cluster 1 : 5, 10, 11, 13, 15

Cluster 2 : 35, 50, 55, 72, 90

Cluster 3 : 204, 215

5. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result:

|       |      |      |      |      |      |      |      |      |      |
|-------|------|------|------|------|------|------|------|------|------|
| age   | 23   | 23   | 27   | 27   | 39   | 41   | 47   | 49   | 50   |
| % fat | 9.5  | 26.5 | 7.8  | 17.8 | 31.4 | 25.9 | 27.4 | 27.2 | 31.2 |
| age   | 52   | 54   | 54   | 56   | 57   | 58   | 58   | 60   | 61   |
| % fat | 34.6 | 42.5 | 28.8 | 33.4 | 30.2 | 34.1 | 32.9 | 41.2 | 35.7 |

- calculate the mean, median and standard deviation of age and % fat.
- Draw the boxplots for age and % fat
- Draw a scatter plot and a q-q plot based on these two variables.
- Normalize the two variables based on z-score normalization.
- calculate the correlation co-efficient).  
Are these two variables positively or negatively correlated.

for age

a. Mean =  $23 + 23 + 27 + 27 + 39 + 41 + 47 + 49 + 50$   
 $+ 52 + 54 + 54 + 56 + 57 + 58 + 58 + 60 + 61$



$$= 46.44$$

for % fat

$$9.5 + 26.5 + 7.8 + 17.8 + 31.4 + 25.9 + 27.4 + 27.2 + 31.2 + 34.6 + 42.5 + 28.8 + 33.4 + 30.2 + 34.1 + 32.9 + 41.2 + 35.7$$

18

$$= 28.78$$

Median for age

23, 23, 27, 27, 39, 41, 47, 49, (50), 52, 54, 54, 56, 57, 58, 58, 60, 61

$$n = 18$$

$$\text{Median} = \frac{50 + 52}{2}$$

$$\text{Age} = \frac{102}{2} = 51$$

for % fat

7.8, 9.5, 17.8, 25.9, 26.5, 27.2, 27.4, 28.8, 30.2, 31.2, 31.4, 32.9, 33.4, 34.1, 34.6, 35.7, 41.2, 42.5

$$\% \text{ fat} = \frac{30.2 + 31.2}{2}$$

$$= \frac{61.4}{2}$$

$$= 30.7\%$$

standard deviation

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

$\sigma$  = Population standard deviation

$\mu$  = 46.44 (mean)

$N$  = 18

$$\sum (x - \mu)^2 = 2971.11$$

divide by  $N$

$$\frac{2971.11}{18} = 165.06$$

$$\sigma = \sqrt{165.06} = 12.85$$

Population standard deviation of age = 12.85

Population standard deviation of % fat

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

$\mu$  = 28.78

$N$  = 18

$$\mu = \frac{518}{18} = 28.78$$

$$\sum (x - \mu)^2 \approx 1454.82$$

$$\frac{1454.82}{18} = 80.82$$

$$\sigma = \sqrt{80.82} = 8.99$$

Population standard deviation of % fat

8.99



b.

minimum = 23

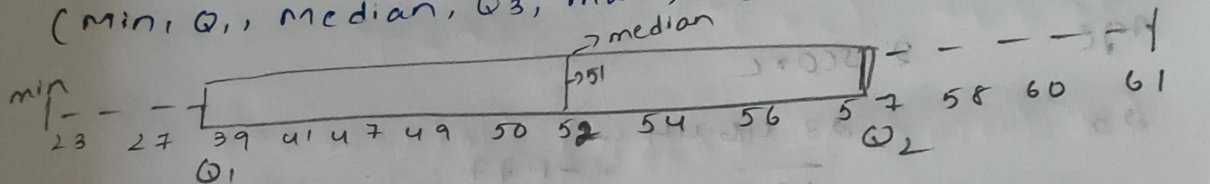
$Q_1$  = median of first 9 values = 39

Median =  $\frac{(50 + 52)}{2} = 51$

$Q_3$  = median of last 9 values = 57

Maximum = 61

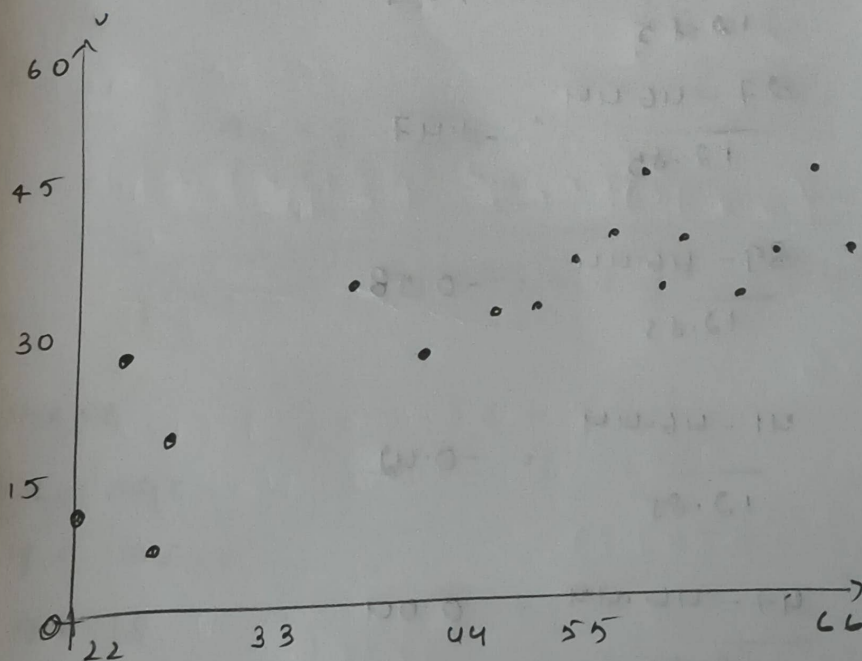
(min,  $Q_1$ , Median,  $Q_3$ , max) = (23, 39, 51, 57, 61)



$gc = 12.85$

at

c. scatter plot



at

# d. Normalization

$$z = \frac{x - \mu}{\sigma}$$

$x$  = original value

$\mu$  = mean = 46.44

$\sigma$  = standard deviation = 12.85

Age

z-score

$$23 \quad \frac{23 - 46.44}{12.85} = -1.77$$

$$23 \quad \frac{23 - 46.44}{12.85} = -1.77$$

$$27 \quad \frac{27 - 46.44}{12.85} = -1.47$$

$$27 \quad \frac{27 - 46.44}{12.85} = -1.47$$

$$39 \quad \frac{39 - 46.44}{12.85} = -0.58$$

$$41 \quad \frac{41 - 46.44}{12.85} = -0.42$$

$$47 \quad \frac{47 - 46.44}{12.85} = 0.04$$

$$49 \quad \frac{49 - 46.44}{12.85} = 0.20$$

$$50 \quad \frac{50 - 46.44}{12.85} = 0.28$$

$$52 \quad \frac{52 - 46.44}{12.85} = 0.43$$



54

$$\frac{54 - 46.44}{12.85} = 0.59$$

56

$$\frac{56 - 46.44}{12.85} = 0.74$$

57

$$\frac{57 - 46.44}{12.85} = 0.82$$

58

$$\frac{58 - 46.44}{12.85} = \cancel{0.89} (0.90)$$

58

$$\frac{58 - 46.44}{12.85} = \cancel{0.87} (0.90)$$

60

$$\frac{60 - 46.44}{12.85} = 1.038$$

61

$$\frac{61 - 46.44}{12.85} = \cancel{1.04} 1.133$$

e.  
d.

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

where

$$x = \text{age}$$

$$y = \% \text{ Fat}$$

$$n = 18$$

$$\sum x = 836$$

$$\sum y = 518.1$$

$$\sum x^2 = \cancel{42584} 41798$$

$$\sum y^2 = 16368.59$$

$$\sum xy = 25763.2$$

$$n \sum xy - \sum x \sum y$$

$$= 18 \times 25763.2 - 836 \times 518.1$$

$$= 463737.6 - 433131.6$$

$$= 30606$$

denominator

|                             |                              |
|-----------------------------|------------------------------|
| $(n \sum x^2 - (\sum x)^2)$ | $(n \sum y^2 - (\sum y)^2)$  |
| $(18(41798) - (836)^2)$     | $(18(16368.59) - (518.1)^2)$ |
| $= 752364 - 698896$         | $= 294634.62 - 268427.61$    |
| $= 53468$                   | $= 26207.1$                  |

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

$$\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}$$

$$= \frac{30606}{\sqrt{(53468)(26207.1)}}$$

$$\sqrt{(53468)(26207.1)}$$

$$= \frac{30606}{\sqrt{1401232462.68}}$$

$$= \frac{30606}{37433.04}$$

$$r = 0.8176$$

$$\boxed{r \approx 0.818} \Rightarrow 0.82$$

As the value  
so, age & fat  
strong positive

similarity

similarity

→ Numerical  
objects are

1. Proximity

2. Contingency

i. Symmetry

d(i)

ii. Association

d(i)

3. Joint

5